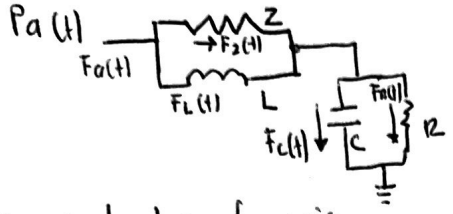
Analisis matematico



Función de transferencia
Ecuaciones principales

$Fa(t) = Fz(t) + FL(t) = Fc(t) + FR(t)$

$Fz(t) = \frac{Pa(t) - Pp(t)}{Z}$

$FL(t) = \frac{1}{L}\int [Pa(t) - Pp(t)]\,dt$

$Fc(t) = C\,\frac{d\,Pp(t)}{dt}$

$FR(t) = \frac{Pp(t)}{R}$

Transformada de la place

$Fa(S) = Fz(S) + FL(S) = Fc(S) + FR(S)$

$Fz(S) = \frac{Pa(S) - Pp(S)}{Z}$

$FL(S) = \frac{Pa(S) - Pp(S)}{LS}$

$Fc(S) = Cs\,Pp(S)$

$FR(S) = \frac{Pp(S)}{R}$

Procedimiento algebraico

$$\frac{Pa(S) - Pp(S)}{Z} + \frac{Pa(S) - Pp(S)}{LS} = Cs\,Pp(S) + \frac{Pp(S)}{R}$$

$$\left(\frac{1}{Z} + \frac{1}{LS}\right) Pa(S) = \left(Cs + \frac{1}{R} + \frac{1}{Z} + \frac{1}{LS}\right) Pp(S)$$

$$\left(\frac{LS + Z}{ZLS}\right) Pa(S) = \left[\frac{CRZLs^2 + ZLs + RLs + RZ}{RZLS}\right] Pp(S)$$

$$\frac{Pp(S)}{Pa(S)} = \frac{\left[\frac{LS+Z}{ZLS}\right]}{\left[\frac{CRZLs^2 + ZLs + RLs + RZ}{RZLS}\right]} = \frac{RLs + RZ}{CRZLs^2 + L(Z+R)s + RZ}$$

Estabilidad en lazo abierto

Normotenso

$Z = 0.01$
$C = 1.5$
$R = 1$
$L = 0.001$

$$\lambda_{1,2} = -\frac{b \pm \sqrt{b^2 - 4ac}}{2a}$$

$a = (0.01)(1.5)(1)(0.001) = 1.5E\text{-}5$

$b = 0.001(0.01 + 1) = 1.01E\text{-}3$

$c = (1)(0.01) = 0.01$

$\lambda_1 = -12.061$

$\lambda_2 = -55.271$

Este sistema presenta una respuesta estable y sobreamortiguada a un escalon unitario de entrada

Hipotenso

$Z = 0.01$
$C = 3.5$
$R = 1$
$L = 0.001$

$a = (0.01)(3.5)(1)(0.001) = 3.5E-5$
$b = 0.001(0.01+1) = 1.01E-3$
$c = (0.01)(1) = 0.01$

$\lambda_1 = -14.428 + 8.805j$
$\lambda_2 = -14.428 - 8.805j$

Este sistema presenta una respuesta estable y subamortiguada a un escalon unitaria

Hipotenso

$Z = 0.01$
$C = 1$
$R = 1$
$L = 0.001$

$a = (0.01)(1)(1)(0.001) = 1E-5$
$b = 0.001(0.01+1) = 1.01E-3$
$c = (0.01)(1) = 0.01$

$\lambda_1 = -11.126$
$\lambda_2 = -89.873$

Este sistema presenta una respuesta estable y sobreamortiguada a un escalon unitario

Error en estado estacionario

$$e(s) = \lim_{s \to 0} s\,Pa(s)\left(1 - \frac{Pp(s)}{Pa(s)}\right) = e(s) = \lim_{s \to 0} \frac{1}{s}\left(1 - \frac{LRs + RZ}{CLRZs^2 + L[R+Z]s + RZ}\right) = 0V$$

$e(s) = 0V$

Modelo de ecuaciones integrodiferenciales

$$F_Z(t) + F_L(t) = F_C(t) + F_R(t)$$

$$\frac{Pa(t) - Pp(t)}{Z} + \frac{1}{L}\int [Pa(t) - Pp(t)]\,dt = C\frac{dPp(t)}{dt} + \frac{Pp(t)}{R}$$

$$\left(\frac{R+Z}{ZR}\right) Pp(t) = \left[\frac{Pa(t)}{Z} + \frac{1}{L}\int [Pa(t) - Pp(t)]\,dt - C\frac{dPp(t)}{dt}\right]$$

$$Pp(t) = \left[\frac{Pa(t)}{Z} + \frac{1}{L}\int [Pa(t) - Pp(t)]\,dt - C\frac{dPp(t)}{dt}\right]\frac{ZR}{Z+R}$$